Generative Artificial Intelligence in Agile Software Development Processes: A Literature Review Focused on User eXperience

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Abstract. The integration of Generative Artificial Intelligence (GenAI) into Agile Software Development (ASD) is reshaping the way development teams automate tasks, optimize processes, and enhance user experience (UX). This study presents a systematic literature review to analyze the impact of GenAI on ASD, focusing on three research questions: (1) How can GenAI tools optimize user experience in agile software development projects? (2) What are agile teams' main challenges when integrating GenAI tools into software development projects? and (3) What stages of the agile software development cycle benefit from implementing GenAI tools? A total of 21 relevant studies published between 2020 and 2024 were selected and analyzed. Findings indicate that GenAI improves UX by facilitating automated test generation, personalized user interfaces, and enhanced documentation processes. However, challenges such as data quality, model transparency, security vulnerabilities, and team resistance hinder its adoption. Moreover, the research highlights that GenAI contributes across multiple ASD phases, including planning (requirement analysis and user story generation), implementation (automated code generation and debugging), testing (self-generated test cases), maintenance (documentation and refactoring), and retrospectives (data-driven team performance analysis). Despite its growing adoption, the study reveals a gap in empirical evaluations of GenAI's long-term impact on Agile methodologies. Future research should explore hybrid frameworks that balance automation and human oversight, longitudinal studies on GenAI's adoption trends, and strategies to ensure ethical and bias-free AI implementation in Agile environments. The findings contribute to a deeper understanding of GenAI's transformative role in software development and provide practical insights for industry professionals and researchers.

A. Coman and S. Vasilache (Eds.): HCII 2025, LNCS 15787, pp. 228–246, 2025.
https://doi.org/10.1007/978-3-031-93536-7_16

Keywords: Generative AI · Agile Software Development · User Experience · Scrum · Software Engineering · Artificial Intelligence

1 Introduction

The integration of GenAI, ASD, and UX represents a key area in modern software engineering. GenAI is revolutionizing various aspects of software development, from code generation to interface design, by producing new and original content based on existing data [15]. Meanwhile, Agile Software Development is a methodology that emphasizes rapid iteration, adaptability, and close collaboration with end users [5,13], making it particularly effective for creating products aligned with user needs. ASD has demonstrated a positive impact on the industry due to its ability to quickly adapt to changes and continuously deliver value. However, it also presents significant challenges, such as maintaining consistency in user experience across multiple iterations and ensuring a cohesive product vision throughout sprints. Other challenges include accurately interpreting user expectations and behaviors, communication overload in multidisciplinary teams, and the limited ability to generate highly personalized solutions in real time. As a result, software products developed under this approach do not always fully meet user demands or require multiple iterations before reaching an acceptable level of functionality and usability [57].

The literature suggests that GenAI models and tools can offer promising solutions to address these challenges by processing complex information and generating structured content. Among the most relevant applications are the automation of user interface prototyping, ensuring consistency with established design principles. Additionally, GenAI can facilitate the translation of user requirements into precise technical specifications, as well as generate design variations that adhere to predefined UX guidelines, thus optimizing the development process.

Despite its potential benefits, Agile teams face critical obstacles that directly impact UX. One of the most significant is ASD's inherent speed, which can lead to inconsistencies in the user interface and overall product experience. As noted by Curcio et al. [8], Agile projects frequently encounter UX design consistency issues during the early development phases. Moreover, effectively capturing and translating user requirements into implementable features remains a persistent challenge [14]. Using GenAI for rapid prototyping and validation of design solutions might compromise UX quality in favor of delivery speed [63]. However, its integration into ASD represents a significant opportunity to enhance UX, as suggested by Li et al. [15]. GenAI tools can significantly optimize development and design processes, allowing teams to focus on more strategic UX design aspects.

Driven by the need to understand how GenAI can impact ASD from a UX-focused perspective, this study addresses the following research questions: (RQ1) How can GenAI tools optimize user experience in Agile software projects? (RQ2) What are the main challenges Agile teams face when integrating GenAI into software development projects? (RQ3) Which specific stages of the Agile software development cycle benefit the most from GenAI implementation? To answer these questions, a systematic literature review was conducted using the PICOC methodology, considering studies indexed in Web of Science, Scopus, IEEE Xplore, and ACM Digital Library. A total of 21 primary studies were selected and analyzed, providing an encouraging perspective on GenAI's role in the evolution of Agile software development.

This paper is organized as follows: Sect. 2 provides the necessary context for the reader to understand the importance, relevance and conceptual bases of the work presented. Section 3 describes the research method used. Section 4 shows the results obtained in the literature review and the evidence found to answer the research questions, and finally, in Sect. 5, the conclusions and future work.

2 Background

In the continuous evolution of software engineering, the integration of generative artificial intelligence with agile development methodologies is offering a new way in which products are conceived and delivered. The capability of GenAI to generate creative content and personalize user interactions, combined with the iterative and collaborative approach of agile practices, lays the foundation for digital solutions more aligned with user expectations and with faster, more effective feedback loops. Along these lines, agile software development has established itself as a flexible and collaborative approach, supported by rapid iterations and constant feedback for the continuous delivery of customer value [17]. Based on the principles of the Agile Manifesto and various methodologies such as Scrum, XP, and Kanban, ASD has demonstrated its effectiveness in the industry by reducing delivery times and increasing adaptability to requirement changes [18]. Nevertheless, the high dynamism of agile teams can pose challenges in maintaining design coherence and user experience in each iteration [19]. The use of agile methodologies has transcended the boundaries of software development, where current literature highlights success stories and applications in fields such as healthcare [21], law [20], and even the fashion industry [22].

UX is a term coined by Norman in 1998 to represent the broader range of human experience quality that extends beyond the traditional concept of usability. While usability focuses on the success of a user's interaction with a product or service, UX encompasses the overall satisfaction and meaning of that interaction [23]. A user-centered approach not only meets functional objectives but also promotes satisfaction and engagement with the product. To this end, the discipline integrates knowledge from psychology, design, and technology, emphasizing cognitive [24,25], emotional [26], and physical interactions, as well as the importance of iterative feedback processes. These principles are applied in domains such as healthcare, e-commerce, and mobile technology [27], highlighting the need for collaboration between academia and industry to close implementation gaps [28]. However, given the complexity and diversity of user experiences, challenges persist in standardizing practices and definitions. At the same time, the growing demand for UX professionals has driven specialized training and the inclusion of courses in curricula, reinforcing an interdisciplinary approach and promoting the effective adoption of UX in practice. Integrating UX into ASD remains a challenge, mainly due to the need to balance the speed of sprints with the consistency and depth of user research.

Generative artificial intelligence refers to systems capable of producing original content (text, images, code, prototypes, etc.) based on patterns learned from large datasets [29,30]. This type of AI differs from deterministic rule-based models. It opens the door to more flexible and personalized interactions and solutions, with applications ranging from healthcare—through the generation of virtual medical images and diagnostic support [31]—to education, commerce, and the preservation of cultural heritage [32,33]. In software engineering, GenAI supports multiple tasks, from code and documentation generation to interface prototyping and user feedback analysis. However, its growing reach raises important ethical and practical considerations, such as bias, the need for explainability, and the implications for strategic decision-making [34,35], requiring careful and responsible integration across diverse domains.

Likewise, the adoption of GenAI in agile frameworks contributes to project management, as its ability to process large volumes of data and provide recommendations based on predictive analysis facilitates decision-making and enhances efficiency in sprint planning, progress monitoring, and team communication [36]. This is particularly relevant for companies immersed in agile transformations since GenAI strengthens creativity and the automation of repetitive tasks, enabling more agile and efficient development cycles [37,38]. By automating routine tasks and analyzing real-time data, organizations can align their agile transformations with the evolution of AI, reinforcing collaboration and keeping the team in a "build-inspect-adapt" cycle [39]. GenAI with non-deterministic models makes UX experiences more dynamic and adaptive, requiring UX professionals to adjust their design strategies to address the opportunities and the challenges arising from greater complexity and personalization in interactions. In UX design processes, there has been an observed improvement in productivity and innovation [36,40]. By automating tasks related to user research and the creation of wireframes, prototype generation, and interface design are optimized, enabling rapid iteration scenarios and the exploration of different design concepts [41]. This user-centered approach, combined with GenAI's ability to respond to user inputs in real-time, drives conversational journeys and more engaging experiences. Generative AI has made a strong entrance into software development by offering advantages in productivity and creativity. In [42], it is reported that tools such as GitHub Copilot and ChatGPT significantly improve programming speed by generating code from natural language. Meanwhile, in [43], the automation of code generation from software requirements is highlighted, alleviating repetitive work and allowing developers to focus on more complex activities. Furthermore, in [44], it is emphasized that large language models (LLM) can directly translate informal descriptions into executable code, streamlining the development process. The "model-based development" approach is also strengthened; as early as 2010 in [45], it is described how this approach facilitates the generation of test cases and models from domain-specific languages (DSL) and UML, contributing to improved software quality and has now been enhanced thanks to generative artificial intelligence, as presented in [46], with applied use in web-based developments [47]. At the application level, it has positioned itself as a transversal technology whose applications span multiple areas and domains. In the medical field, image translation (for example, converting T1 to T2 scans) improves diagnostic accuracy and obtain sharper images, even in unpaired translation scenarios [48]. Also, models have been developed to transform SAR images into high-resolution optical images, which is valuable for satellite observation and remote sensing [48]. In editing and style, generative networks allow for modifications of facial features or the transfer of artistic styles. At the same time, in education, they facilitate the production of synthetic material and multilingual support [48]. Moreover, methods such as BERT and ELMo have proven effective for classification and text-mining tasks, encompassing even malware detection and summary synthesis [48]. Finally, video generation and editing techniques add a dynamic layer to these capabilities, broadening the range of content that can be created or transformed automatically [48].

Generative AI increases the accuracy and quality of results in specialized applications and drives innovation and efficiency across numerous processes. Nevertheless, certain limitations persist, such as the need for more excellent reliability in the models, the interpretability of their results, and the consideration of ethical implications. Despite these challenges, its transformative capacity positions it as a research area still far from reaching its ceiling, with a potential that promises to continue expanding the frontiers of software engineering and other technological fields.

3 Methodology

The main goal of this work is to analyze, through a comprehensive literature review, how GenAI tools can optimize UX in agile software development projects. To achieve this objective, a literature review was conducted based on the guidelines proposed by [52,60]. The process was conducted through the following steps: 1) statement of research questions, 2) search process, 3) selection of studies, and 4) analysis of results. All these steps are described below.

3.1 Research Questions

We have defined three specific research questions for more detailed knowledge and a comprehensive view of the subject. The research questions to be answered in this study are as follows:

- RQ1. How can GenAI tools optimize user experience in agile software development projects?
- RQ2. What are agile teams' main challenges when integrating GenAI tools into software development projects?

– RQ3. What stages of the agile software development cycle benefit from implementing GenAI tools?

3.2 Search Process

The citation databases used were SCOPUS and Web of Science (WoS), while the scientific publication databases used were ACM Digital Library and IEEE Xplore. This selection is mainly due to the reputation of these databases in the discipline and the fact that we have full access to the published material. To construct the search string, we use the method PICOC [12,52,59] whose acronyms represent the criteria that drive the process. The PICOC method is structured to formulate research questions and construct efficient search chains. PICOC is an acronym that stands for five key elements: **P** de Population; **I** de Intervention; **C** de Comparison; **O** de Outcomes y **C** de Context. Based on the PICOC method, the following search string was developed: *(organization OR universities OR educational OR company OR Institution) AND (GenAI OR GAI OR "Generative Artificial Intelligence" OR AI OR "artificial intelligence") AND (agile OR scrum OR agileRE OR "agile requirements engineering") AND (tools OR software OR program OR application) AND (limitations OR guidelines OR implementation OR challenge)*. This search string was validated with a set of papers we identified and used as a control group. The investigation was initially carried out in November; its last update was mid-December 2024.

The inclusion/exclusion criteria were as follows:

– Published between 2020 and 2024.
– Contain information on GenAI and Agile Software Development.
– Only written in English.

3.3 Selection of Primary Studies

The data extraction form was developed with the following fields: paper Title, Year, DOI; Published in (Journal, Conference); Study Type (Quantitative, Qualitative, Mixed); Research Method (Literature Review, Case Study, Interviews, Questionnaires, Observation); Objective; Main Findings; Main Conclusions; Future Work; Tributes to RQ1 (Yes/No); Evidence RQ1; Tributes to RQ2 (Yes/No); Evidence RQ2; Tributes to RQ3 (Yes/No); Evidence RQ3; Number of people; Number of Women; Number of Men.

The methodology for searching and selecting papers was structured into three successive stages:

– **First Stage (1F):** We began by reviewing all titles and keywords of the papers obtained from each database. Subsequently, duplicates were removed to ensure the uniqueness of the documents.
– **Second Stage (2F):** Next, we conducted a detailed reading of the abstracts of those papers that passed the first filter. This step allowed a preliminary assessment of the relevance and pertinence of each paper.

– **Third Stage:** Finally, the papers selected after the second filter were downloaded, read in their entirety, and recorded in a form designed in Microsoft Excel, following the established protocol. This stage involved an exhaustive review to determine their final inclusion in the study.

After executing the search string in each consulted database, we obtained the following initial results: 92 papers from Web of Science (WoS), 26 from SCOPUS, 41 from IEEE Xplore, and 261 from ACM Digital Library. Subsequently, we applied the previously described filters (1F and 2F), which resulted in a more refined selection: 3 papers from WoS, 2 from SCOPUS, 5 from IEEE Xplore, and 11 from ACM Digital Library. Finally, we selected 21 primary studies for detailed analysis and discussion.(see Table 1).

3.4 Analysis of Results

The analysis of the selected primary studies reveals several key trends in GenAI research within Agile Software Development. In terms of publication years, there is a notable concentration of studies in 2024, highlighting the growing interest of the scientific community in the intersection of GenAI, Agile methodologies, and software development. This trend suggests that research in this field is experiencing a phase of rapid expansion, driven by recent advancements in technologies such as large language models (LLMs) and their integration into development workflows. Additionally, this surge in interest aligns with the accelerated digital transformation in key industries, which has increased the demand for innovative solutions to optimize development processes and enhance user experience.

The analyzed studies were published in highly recognized academic and professional sources, including *IEEE Access*, *ACM Transactions*, and *MDPI Information*, as well as high-impact specialized conferences. The presence of these works in prestigious journals and conferences not only validates their methodological quality and scientific relevance but also reflects a multidisciplinary approach, combining theoretical, empirical, and applied contributions. This balance ensures that the findings are valuable for both the academic community and industry professionals, facilitating knowledge transfer between research and practice.

Regarding the application context, most studies focus on business and academic environments, particularly in medium and large enterprises, as well as in educational and technological institutions. This highlights the dual nature of research, bridging real-world application and conceptual development within academia, demonstrating a growing interest in exploring the applicability of GenAI in Agile projects across different domains. Finally, the results identify key challenges that warrant further investigation, such as organizational resistance to change, lack of transparency in AI models, and technological interoperability in heterogeneous development environments. These findings open new research avenues, including the customization of GenAI tools for specific contexts and the improvement of data quality to enhance the integration of these technologies into Agile software development.

Table 1. Selected primary studies (PS) and DOI

ID	Title	DOI
[PS_1]	The Potential of AI-Driven Assistants in Scaled Agile Software Development [62]	10.3390/app14010319
[PS_2]	A Roadmap for Software Testing in Open-Collaborative and AI-Powered Era [54]	10.1145/3709355
[PS_3]	An Agile New Research Framework for Hybrid Human-AI Teaming: Trust, Transparency, and Transferability [56]	10.1145/3514257
[PS_4]	Applying Machine Learning to Estimate the Effort and Duration of Individual Tasks in Software Projects [4]	10.1109/ACCESS.2023.3307310
[PS_5]	Artificial Intelligence and Agility-Based Model for Successful Project Implementation and Company Competitiveness [53]	10.3390/info14060337
[PS_6]	Beyond Code Generation: An Observational Study of ChatGPT Usage in Software Engineering Practice [55]	10.1145/3660788
[PS_7]	GenAI and Software Engineering: Strategies for Shaping the Core of Tomorrow's Software Engineering Practice [6]	10.14742/ajet.8922
[PS_8]	FRAMUX-EV: A Framework for Evaluating User Experience in Agile Software Development [16]	10.3390/app14198991
[PS_9]	Comprehensive Evaluation and Insights Into the Use of Large Language Models in the Automation of Behavior-Driven Development Acceptance Test Formulation [58]	10.1109/ACCESS.2024.3391815
[PS_10]	Cypress Copilot: Development of an AI Assistant for Boosting Productivity and Transforming Web Application Testing [61]	10.1109/ACCESS.2024.3521407
[PS_11]	Development in Times of Hype: How Freelancers Explore Generative AI? [50]	10.1145/3597503.3639111
[PS_12]	Enhancing educational efficiency: Generative AI chatbots and DevOps in Education 4.0 [10]	10.1002/cae.22804
[PS_13]	Estimating Software Functional Size via Machine Learning [49]	10.1145/3582575
[PS_14]	From Today's Code to Tomorrow's Symphony: The AI Transformation of Developer's Routine by 2030 [7]	10.1145/3709353
[PS_15]	From Triumph to Uncertainty: The Journey of Software Engineering in the AI Era [3]	10.1145/3709360
[PS_16]	How to Write Ethical User Stories? Impacts of the ECCOLA Method [11]	10.1007/978-3-030-78098-2_3
[PS_17]	Human-AI Collaboration in Software Engineering: Lessons Learned from a Hands-On Workshop [51]	10.1145/3643690.3648236
[PS_18]	Integrating Generative AI for Advancing Agile Software Development and Mitigating Project Management Challenges [2]	10.14569/IJACSA.2024.0150306
[PS_19]	Large Language Models for Software Engineering: A Systematic Literature Review [64]	10.1145/3695988
[PS_20]	Requirements are All You Need: The Final Frontier for End-User Software Engineering [9]	10.1145/3708524
[PS_21]	Using Voice and Biofeedback to Predict User Engagement during Product Feedback Interviews [1]	10.1145/3635712

On the other hand, Table 2 offers an integrated and organized view of how the selected primary studies align with the previously defined research questions. This table shows the correspondence between the questions and the studies and facilitates a deeper understanding of the focus areas and specific aspects that each study addresses in relation to the posed questions. This provides a valuable tool for quickly identifying the key contributions of each study in the broader context of the research.

Table 2. Matching research questions and primary studies

Research Questions	References to primary studies
RQ1. How can GenAI tools optimize user experience in agile software development projects?	PS_5, PS_7, PS_10, PS_14, PS_17, PS_18, PS_19, PS_20
RQ2. What are agile teams' main challenges when integrating GenAI tools into software development projects?	PS_1, PS_2, PS_3, PS_4, PS_5, PS_6, PS_7, PS_8, PS_9, PS_10, PS_11, PS_12, PS_13, PS_14, PS_15, PS_16, PS_17, PS_18, PS_19, PS_20, PS_21
RQ3. What stages of the agile software development cycle benefit from implementing GenAI tools?	PS_1, PS_2, PS_3, PS_4, PS_5, PS_6, PS_7, PS_8, PS_9, PS_10, PS_11, PS_12, PS_13, PS_14, PS_15, PS_16, PS_17, PS_18, PS_19, PS_20, PS_21

3.5 Limitations

While this systematic literature review provides a detailed perspective on the impact of GenAI in ASD and its relationship with UX, several limitations must be considered when interpreting the findings.

First, the selection of studies was restricted to publications indexed in Web of Science, Scopus, IEEE Xplore, and ACM Digital Library, which, while ensuring high academic rigor and quality, excluded gray literature and industry sources, such as technical reports, case studies in private companies, and documentation of commercial tools. This may have limited the representation of practical, industry-driven approaches, particularly regarding the real-world application of GenAI in software development environments.

Second, the study focused on publications from 2020 to 2024, reflecting the early stage of GenAI adoption in ASD. As a result, many analyzed studies present preliminary or short-term evaluations, making it difficult to assess long-term trends and the sustained impact of these technologies on team productivity, UX evolution, and organizational readiness for GenAI integration.

Another critical limitation is the lack of uniformity in the evaluation methods for GenAI in ASD. The reviewed studies employ diverse methodological approaches, including controlled experiments, qualitative analyses, and literature reviews, but few use standardized metrics to assess the impact of GenAI on the software lifecycle. This variability complicates direct comparisons between findings and highlights the need for more robust and replicable methodological frameworks in future research.

Lastly, this review focused on studies discussing GenAI in a general Agile development context, without distinguishing differences between specific Agile methodologies (Scrum, Kanban, XP, SAFe). This raises the possibility that the applicability and effectiveness of GenAI may vary depending on the Agile framework used. Future research should explore the integration of GenAI within specific Agile methodologies, assessing tailored implementation strategies.

Despite these limitations, the findings provide a solid and up-to-date perspective on the intersection of GenAI, ASD, and UX, serving as a relevant foundation for further exploration. Future studies are encouraged to expand the range of considered sources, incorporate longitudinal analyses, and develop standardized methodologies to objectively evaluate the impact of GenAI on Agile software development.

4 Discussion of Results

As described in Sect. 4, a total of 21 articles were selected as primary studies and subsequently analyzed and classified based on their alignment with the research questions that guided this study. This process allowed us to identify key information and extract relevant evidence to address each of the formulated research questions. The following section presents a synthesis of the findings and their relationship with the research questions.

4.1 RQ1: How Can GenAI Tools Optimize User Experience in Agile Software Development Projects?

GenAI tools, such as ChatGPT, GitHub Copilot, and other advanced language models, have proven to be a key resource for optimizing user experience in Agile projects. Their impact is evident in both the improvement of internal development processes and the quality of the final products delivered to users. These tools enable automation of repetitive tasks, technical documentation generation, and tailored solutions for complex problems, resulting in greater efficiency in software development and an enhanced user experience.

Studies [2,6,7,9,51,53,61,64] highlight that the automation of tasks such as code generation, testing, and documentation allows development teams to focus on higher-value activities, leading to more robust software that aligns with customer needs. Additionally, tools like ChatGPT enhance team collaboration and data-driven decision-making, promoting a more personalized user experience.

In particular, studies [9,51] emphasize how GenAI is transforming traditional workflows in Agile environments. Its ability to automatically generate prototypes and functionalities from natural language requirements reduces barriers between users and development teams, fostering greater client participation in the product creation process.

A notable study by [53] evaluated the impact of ChatGPT on developer efficiency in prototype creation. Through an experimental setup, it was observed that developers using ChatGPT reduced development time by 40% and delivered prototypes more closely aligned with customer requirements, ultimately improving the user experience. Similarly, the study by [6] examines how GitHub

Copilot optimizes workflows, highlighting that developers were able to generate code faster and reduce common errors, allowing them to focus on enhancing the user experience.

Furthermore, the study by [61] explores how generative language models can personalize interactions and functionalities from the early stages of development. Through mixed experiments and surveys, it was found that 85% of end users reported significant improvements in their experience, mainly due to the creation of more intuitive and user-friendly interfaces.

Additionally, the study by [7] highlights the impact of tools like Tabnine in automating real-time technical documentation generation, which reduced user training time and improved comprehension of the final product. Finally, the study by [51] describes how GenAI facilitates active customer participation in the development process, enabling them to translate their requirements into functional prototypes via natural language interfaces, ensuring products that better align with their expectations.

4.2 RQ2: What Are Agile Teams' Main Challenges When Integrating GenAI Tools Into Software Development Projects?

The integration of GenAI tools in Agile projects presents a series of technical, organizational, and ethical challenges that require significant adaptation within development teams. The use of GenAI tools in software implementation has significantly optimized code generation, reducing development time and minimizing common errors. However, since AI models can produce code with bugs or vulnerabilities, it is crucial to integrate automated validation processes. The analyzed studies consistently identify issues related to data quality, trust in AI tools, and the need for cultural adaptation within Agile teams.

One of the most recurrent challenges, identified in studies [7,9,51,53,61], is data quality. GenAI tools, such as large language models (LLMs), rely on well-structured and representative datasets to generate accurate results. However, inadequate or biased training data can introduce errors that negatively impact development processes. According to [55], biases in training data affect the accuracy and fairness of GenAI outputs, leading to software implementation failures and ultimately compromising product quality.

Another significant challenge, highlighted in [16], is the technical limitations of GenAI models, such as generating hallucinations or incorrect responses, which necessitate an intensive manual validation process. This increases the workload for Agile teams, reducing the expected benefits of automation. Additionally, studies [50] address the security risks associated with tools like GitHub Copilot, which may generate code with vulnerabilities. To mitigate these risks, teams must implement additional review and quality control processes, adding complexity and costs to software projects.

Another notable obstacle, discussed in [49], is the lack of explainability in GenAI models. These models often function as black boxes, making it difficult to adopt them in environments where transparency is crucial, particularly in regulated industries or projects requiring decision traceability. This limitation restricts their implementation in contexts that demand clear justifications for AI-driven decisions.

On an organizational level, resistance to change and lack of technical training within Agile teams pose a critical challenge, as described in [11]. Many developers remain skeptical about the impact of GenAI on their work, fearing a loss of control or relevance in the development process [51,61]. This resistance can slow down the adoption of these tools, reducing their effectiveness and creating additional barriers to their integration into workflows. To address this issue, it is essential to develop strategies that promote training, controlled experimentation, and human-AI collaboration within development teams.

Finally, Table 3 groups the main challenges identified in the primary studies, providing a structured summary of the barriers to GenAI integration in Agile projects.

Table 3. Challenges of Integrating GenAI in Agile Teams

Identified Challenge	Description	Primary Studies
Data Quality	GenAI models rely on well-structured and representative data to generate accurate results. If the data is biased or of low quality, it may lead to incorrect or skewed information. **Example:** Language models generating code with accessibility biases.	PS_6, PS_8, PS_9, PS_11, PS_13, PS_16
Technical Limitations	GenAI struggles with solving complex problems, often generating incorrect answers or hallucinations. This requires developers to manually verify the results. **Example:** Models unable to handle advanced logic in code.	PS_8, PS_10, PS_11, PS_13, PS_14
Security Risks	Tools like GitHub Copilot may generate code with vulnerabilities, increasing the risk of cyberattacks. **Example:** Generated code with authentication flaws or exposure of sensitive data.	PS_7, PS_10, PS_11, PS_19, PS_20
Lack of Explainability	GenAI models function as "black boxes," making it difficult to understand how they generate results. This can be problematic in environments where traceability is crucial. **Example:** Models recommending code changes without justifying the underlying logic.	PS_13, PS_14, PS_16, PS_19
Resistance to Change	Developers may be reluctant to adopt GenAI due to a lack of trust in its capabilities or fear of losing control over the code. **Example:** Teams preferring to write code manually rather than accepting AI-generated suggestions.	PS_10, PS_11, PS_16, PS_17
Excessive Dependence on GenAI	Overreliance on GenAI may lead to dependency, reducing the development of technical skills within teams. **Example:** Developers relying on GenAI to write code without understanding the underlying logic.	PS_7, PS_12, PS_15, PS_18

4.3 RQ3: What Stages of the Agile Software Development Cycle Benefit from Implementing GenAI Tools?

GenAI tools play a significant role across multiple stages of the Agile development cycle, from planning to maintenance. Their integration into development processes enhances efficiency through task automation, predictive analytics, and real-time documentation, ultimately improving team performance and software quality.

According to [62], in the planning stage, tools like ChatGPT assist in generating user stories and prioritizing tasks based on natural language requirements. This improves the accuracy of sprint goal definition, ensuring better alignment between user needs and the developed functionalities.

During the implementation phase, the study by [54] explains how GenAI automates the generation of optimized and real-time reviewed code, significantly reducing developers' manual workload and minimizing errors in the coding process.

In the testing phase, [10] describe how these tools enable automated test case generation and early error detection, thereby increasing the reliability and stability of the final product.

For maintenance, the study by [3] highlights the role of tools like Tabnine in continuously updating technical documentation, which not only ensures software scalability but also facilitates the onboarding of new developers into projects.

Finally, in the retrospective phase, [1] explain how GenAI helps analyze team performance metrics and generate data-driven recommendations to enhance future Agile iterations, optimizing team dynamics and overall development efficiency.

Table 4 presents the Agile development stages identified in the primary studies, along with the corresponding GenAI tools applied in each phase.

Table 4. Agile Development Stages and Identified GenAI Tools

Agile Development Stage	Identified GenAI Tools	Primary Studies
Planning	ChatGPT for user story generation and requirement analysis; LLM models for effort estimation and sprint planning.	PS_1, PS_2, PS_4, PS_12, PS_15
Implementation	GitHub Copilot and Tabnine for code generation, real-time error detection, and code optimization.	PS_2, PS_5, PS_10, PS_15, PS_19
Testing	ChatGPT and specialized testing models for automated test case generation, error analysis, and code coverage.	PS_3, PS_10, PS_14, PS_18, PS_20
Maintenance	Tabnine and automatic documentation models for continuous technical documentation updates and code refactoring.	PS_6, PS_11, PS_16, PS_17, PS_21
Retrospectives	Predictive analysis models and ChatGPT for team performance metric analysis and improvement recommendations.	PS_7, PS_9, PS_13, PS_17, PS_20

The increasing use of GenAI tools in Agile software development presents both a strategic opportunity and a technical and organizational challenge that must be addressed holistically. While previous studies have highlighted productivity improvements and task automation, the findings of this research suggest that effectively integrating GenAI into Agile teams requires specific strategies to mitigate risks related to data quality, code security, and model transparency.

Furthermore, existing literature has primarily focused on the benefits of GenAI during the implementation phase, leaving a gap in understanding its impact on other critical Agile stages, such as planning, maintenance, and retrospectives. This study helps bridge that gap by demonstrating that GenAI also facilitates continuous software improvement through predictive analysis of performance metrics and technical documentation optimization, enabling a more seamless integration of these technologies into Agile development.

For a sustainable adoption of GenAI in Agile environments, it is essential to develop hybrid frameworks that combine the automation capabilities of GenAI with human oversight practices, ensuring both operational efficiency and decision traceability within development processes. This approach will not only maximize the benefits of GenAI but also mitigate associated risks, ensuring its ethical and effective implementation in Agile teams.

5 Conclusions and Future Work

The findings of this research have demonstrated that the integration of GenAI tools into ASD provides significant benefits in terms of optimizing user experience, reducing development errors, and automating repetitive tasks. However, several critical challenges have also been identified, including data quality, model opacity, resistance to change within teams, and code security.

Regarding RQ1, the analyzed studies suggest that GenAI enhances user experience by enabling interface personalization, usability test automation, and prototype development optimization. Additionally, it facilitates more efficient interactions between development teams and end users, reducing barriers in requirement capture and translation, ultimately leading to products that better align with user expectations.

With respect to RQ2, the main challenges identified include the need to ensure data quality and representativeness, avoid defective or vulnerable code generation, and ensure transparency in AI-driven decisions. Additionally, some development teams have shown resistance to adopting these tools, highlighting the importance of training strategies and progressive adaptation to foster effective integration.

Regarding RQ3, the reviewed studies indicate that GenAI positively impacts multiple stages of Agile development. During the planning phase, it facilitates user story creation and task prioritization. In the implementation phase, it optimizes code generation and reduces real-time errors. During testing, it automates the generation and execution of test cases, while in the maintenance stage, it improves technical documentation and code refactoring. Finally, in the retrospective phase, it enables a more precise analysis of performance metrics, generating data-driven recommendations that support continuous process improvement.

Despite these advancements, research in this field remains in an early stage, and several opportunities for future studies have been identified. First, a more in-depth empirical evaluation is needed to assess GenAI's impact on Agile team productivity and software quality. Additionally, it is important to analyze how Agile methodologies can be adapted to maximize GenAI's benefits, considering factors such as organizational culture and team technological maturity. Another promising research avenue involves developing hybrid frameworks that combine GenAI with tailored Agile approaches based on the type of software being developed. Furthermore, future studies could explore methodologies to ensure ethical and transparent use of GenAI in Agile processes, as well as strategies to mitigate inherent biases in generative language models, ensuring a more inclusive and reliable software development process.

Acknowledgments. This work was partially funded by DIUDA (Dirección de Investigación de la Universidad de Atacama) Regular Project of the Universidad de Atacama, No. 22386.

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